EXPERIMENT 2

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Semester: 5
Subject Name: Computer Network **Subject Code: 24CSH-337**

Experiment 02 – Basic Network Commands and Network Configuration

Aim

To study and execute various Windows networking commands used for configuring, monitoring, and troubleshooting computer networks.

Objectives

- To understand the purpose of common networking commands.
- To verify network connectivity using Ping.
- To examine network configuration using IPConfig.
- To trace packet routes using Tracert.
- To perform DNS lookups using NSLookup.
- To monitor active network connections using Netstat.
- To view ARP cache entries and IP-to-MAC mappings.
- To identify the system hostname.
- To analyze packet loss using PathPing.
- To study routing information and shared network resources using Route and Net Use.

Software Requirements

Software

- Microsoft Windows 11
- Command Prompt (CMD)

Hardware

- Desktop/Laptop

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- Network Interface Card (NIC)
- Internet or LAN Connection

Procedure

1. Open Command Prompt with appropriate permissions.
2. Execute the Ping command and its commonly used variations.
3. Display the system's IP configuration using IPConfig commands.
4. Trace the route to a destination using Tracert.
5. Perform DNS lookups using NSLookup.
6. Display active network connections using Netstat.
7. View the ARP cache using the ARP command.
8. Display the computer name using the Hostname command.
9. Analyze network latency and packet loss using PathPing.
10. View the routing table using the Route command.
11. Display network share information using the Net Use command.
12. Observe and record the outputs of each command.

Commands Executed

Ping

ping google.com

ping -t google.com

ping -n 5 google.com

ping -l 1000 google.com

ping -f google.com

ping -a 8.8.8.8

IPConfig

ipconfig

ipconfig /all

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ipconfig /release

ipconfig /renew

Tracert

tracert google.com

tracert -d google.com

tracert -h 10 google.com

NSLookup

nslookup google.com

nslookup 8.8.8.8

nslookup google.com 8.8.8.8

nslookup -type=A google.com

Netstat

netstat

netstat -a

netstat -n

netstat -o

netstat -ano

ARP

arp -a

arp -d *

Hostname

hostname

PathPing

pathping google.com

Route

route print

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Net Use

net use

Step-wise Output

Step 1 – Ping Command

Step 2 – Ping Variations

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```
PS C:\Users\HP> ping -t google.com

Pinging google.com [2404:6800:4002:81c::200e] with 32 bytes of data:
Reply from 2404:6800:4002:81c::200e: time=48ms
Reply from 2404:6800:4002:81c::200e: time=41ms
Reply from 2404:6800:4002:81c::200e: time=49ms
Reply from 2404:6800:4002:81c::200e: time=70ms
Reply from 2404:6800:4002:81c::200e: time=55ms
Reply from 2404:6800:4002:81c::200e: time=76ms

Ping statistics for 2404:6800:4002:81c::200e:
    Packets: Sent = 6, Received = 6, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 41ms, Maximum = 76ms, Average = 56ms
Control-C
PS C:\Users\HP>
```

```
PS C:\Users\HP> ping -n 5 google.com

Pinging google.com [2404:6800:4002:81c::200e] with 32 bytes of data:
Reply from 2404:6800:4002:81c::200e: time=62ms
Reply from 2404:6800:4002:81c::200e: time=68ms
Reply from 2404:6800:4002:81c::200e: time=46ms
Reply from 2404:6800:4002:81c::200e: time=42ms
Reply from 2404:6800:4002:81c::200e: time=62ms

Ping statistics for 2404:6800:4002:81c::200e:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 42ms, Maximum = 68ms, Average = 56ms
PS C:\Users\HP>
```

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```
PS C:\Users\HP> ping -l 100 google.com

Pinging google.com [2404:6800:4002:81b::200e] with 100 bytes of data:
Reply from 2404:6800:4002:81b::200e: time=78ms
Reply from 2404:6800:4002:81b::200e: time=83ms
Reply from 2404:6800:4002:81b::200e: time=91ms
Reply from 2404:6800:4002:81b::200e: time=86ms

Ping statistics for 2404:6800:4002:81b::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 78ms, Maximum = 91ms, Average = 84ms
PS C:\Users\HP>
```

```
PS C:\Users\HP> ping -f -l 1472 google.com

Pinging google.com [142.250.182.206] with 1472 bytes of data:
Reply from 10.87.27.170: Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.

Ping statistics for 142.250.182.206:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
PS C:\Users\HP>
```

Step 3 – IPConfig

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```
C:\Users\Utkar>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4
   Temporary IPv6 Address. . . . . . : 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05
   Link-local IPv6 Address . . . . . : fe80::948e:7197:14dc:d497%19
   IPv4 Address. . . . . . . . . . . : 192.168.1.31
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%19
                                       192.168.1.1
```

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```
C:\Users\Utkar>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Utkarsh
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : AC-19-8E-E8-F5-F4
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : AE-19-8E-E8-F5-F3
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
   Physical Address. . . . . . . . . : B2-4A-34-58-0C-C4
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4(Preferred)
   Temporary IPv6 Address. . . . . . : 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05(Preferred)
   Link-local IPv6 Address . . . . . : fe80::948e:7197:14dc:d497%19(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.31(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 17 July 2026 20:09:07
   Lease Expires . . . . . . . . . . : 18 July 2026 20:11:44
   Default Gateway . . . . . . . . . : fe80::1%19
                                       192.168.1.1
```

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```
C:\Users\Utkar>ipconfig/release

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4
   Temporary IPv6 Address. . . . . . : 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05
   Link-local IPv6 Address . . . . . : fe80::948e:7197:14dc:d497%19
   Default Gateway . . . . . . . . . : fe80::1%19
```

```
C:\Users\Utkar>ipconfig /renew

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4
   Temporary IPv6 Address. . . . . . : 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05
   Link-local IPv6 Address . . . . . : fe80::948e:7197:14dc:d497%19
   IPv4 Address. . . . . . . . . . . : 192.168.1.53
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%19
                                       192.168.1.1
```

Step 4 – Tracert

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```
C:\Users\Utkar>tracert python-quiz.pages.dev

Tracing route to python-quiz.pages.dev [2606:4700:310c::ac42:2d23]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2
  2     *        *        *     Request timed out.
  3     3 ms     3 ms     2 ms  2404:a800:1a00:804::cd
  4    13 ms    13 ms    14 ms  2404:a800:88::134
  5    15 ms    15 ms     *     2404:a800:0:29::cf
  6    13 ms    13 ms    17 ms  2606:4700:310c::ac42:2d23

Trace complete.
```

```
C:\Users\Utkar>tracert -d python-quiz.pages.dev

Tracing route to python-quiz.pages.dev [2606:4700:310c::ac42:2d23]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2
  2     *        *        *     Request timed out.
  3     2 ms     2 ms     2 ms  2404:a800:1a00:804::cd
  4    13 ms    13 ms    13 ms  2404:a800:88::134
  5     *       15 ms    15 ms  2404:a800:0:29::cf
  6    13 ms    11 ms    11 ms  2606:4700:310c::ac42:2d23

Trace complete.
```

```
C:\Users\Utkar>tracert -h 5 python-quiz.pages.dev

Tracing route to python-quiz.pages.dev [2606:4700:310c::ac42:2edd]
over a maximum of 5 hops:

  1     1 ms     1 ms     1 ms  2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2
  2     *        *        *     Request timed out.
  3     3 ms     2 ms     2 ms  2404:a800:1a00:804::cd
  4    11 ms    13 ms    11 ms  2404:a800:88::134
  5    12 ms    13 ms    12 ms  2404:a800:0:29::cf

Trace complete.
```

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```
C:\Users\Utkar>tracert -?

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.
```

Step 5 – NSLookup

```
C:\Users\Utkar>nslookup python-quiz.pages.dev
Server:  192.168.1.1
Address:  192.168.1.1

Non-authoritative answer:
Name:    python-quiz.pages.dev
Addresses:  2606:4700:310c::ac42:2edd
          2606:4700:310c::ac42:2d23
          172.66.46.221
          172.66.45.35
```

```
C:\Users\Utkar>nslookup -type=A python-quiz.pages.dev
Server:  192.168.1.1
Address:  192.168.1.1

Non-authoritative answer:
Name:    python-quiz.pages.dev
Addresses:  172.66.45.35
          172.66.46.221
```

Step 6 – Netstat

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```
C:\Users\Utkar>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49367        Utkarsh:64463          ESTABLISHED
  TCP    127.0.0.1:49696        Utkarsh:49697          ESTABLISHED
  TCP    127.0.0.1:49697        Utkarsh:49696          ESTABLISHED
  TCP    127.0.0.1:49699        Utkarsh:49700          ESTABLISHED
  TCP    127.0.0.1:49700        Utkarsh:49699          ESTABLISHED
  TCP    127.0.0.1:49707        Utkarsh:49708          ESTABLISHED
  TCP    127.0.0.1:49708        Utkarsh:49707          ESTABLISHED
  TCP    127.0.0.1:56058        Utkarsh:56059          ESTABLISHED
  TCP    127.0.0.1:56059        Utkarsh:56058          ESTABLISHED
  TCP    127.0.0.1:63101        Utkarsh:63102          ESTABLISHED
  TCP    127.0.0.1:63102        Utkarsh:63101          ESTABLISHED
  TCP    127.0.0.1:64076        Utkarsh:64464          ESTABLISHED
  TCP    127.0.0.1:64463        Utkarsh:49367          ESTABLISHED
  TCP    127.0.0.1:64464        Utkarsh:64076          ESTABLISHED
  TCP    192.168.1.53:50647     192:8009               ESTABLISHED
  TCP    192.168.1.53:50648     ec2-107-20-21-3:https  ESTABLISHED
  TCP    192.168.1.53:50649     192:8009               ESTABLISHED
  TCP    192.168.1.53:52688     ec2-52-45-140-188:https  ESTABLISHED
  TCP    192.168.1.53:54189     192:8009               ESTABLISHED
  TCP    192.168.1.53:56093     192:8009               ESTABLISHED
  TCP    192.168.1.53:56094     192:8009               ESTABLISHED
  TCP    192.168.1.53:56098     192:8009               ESTABLISHED
  TCP    192.168.1.53:56522     192:8009               ESTABLISHED
  TCP    192.168.1.53:56600     ec2-18-97-36-65:https  ESTABLISHED
  TCP    192.168.1.53:57783     192:8008               ESTABLISHED
  TCP    192.168.1.53:57784     192:8008               ESTABLISHED
  TCP    192.168.1.53:57789     192:8008               ESTABLISHED
  TCP    192.168.1.53:57790     192:8008               ESTABLISHED
  TCP    192.168.1.53:60739     lb-140-82-113-26-iad:https  ESTABLISHED
  TCP    192.168.1.53:60741     ec2-98-85-208-136:https  ESTABLISHED
  TCP    192.168.1.53:60951     192:8009               ESTABLISHED
  TCP    192.168.1.53:61178     ec2-18-97-36-44:https  ESTABLISHED
```

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```
C:\Users\Utkar>netstat -o

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    127.0.0.1:49367        Utkarsh:64463          ESTABLISHED     11860
  TCP    127.0.0.1:49696        Utkarsh:49697          ESTABLISHED     1912
  TCP    127.0.0.1:49697        Utkarsh:49696          ESTABLISHED     1912
  TCP    127.0.0.1:49699        Utkarsh:49700          ESTABLISHED     3228
  TCP    127.0.0.1:49700        Utkarsh:49699          ESTABLISHED     3228
  TCP    127.0.0.1:49707        Utkarsh:49708          ESTABLISHED     5300
  TCP    127.0.0.1:49708        Utkarsh:49707          ESTABLISHED     5300
  TCP    127.0.0.1:56058        Utkarsh:56059          ESTABLISHED     1372
  TCP    127.0.0.1:56059        Utkarsh:56058          ESTABLISHED     1372
  TCP    127.0.0.1:63101        Utkarsh:63102          ESTABLISHED     26820
  TCP    127.0.0.1:63102        Utkarsh:63101          ESTABLISHED     26820
  TCP    127.0.0.1:64076        Utkarsh:64464          ESTABLISHED     11860
  TCP    127.0.0.1:64463        Utkarsh:49367          ESTABLISHED     12244
  TCP    127.0.0.1:64464        Utkarsh:64076          ESTABLISHED     12244
  TCP    192.168.1.53:50647     192:8009               ESTABLISHED     19924
  TCP    192.168.1.53:50648     ec2-107-20-21-3:https  ESTABLISHED     7172
  TCP    192.168.1.53:50649     192:8009               ESTABLISHED     12244
  TCP    192.168.1.53:52688     ec2-52-45-140-188:https  ESTABLISHED     7172
  TCP    192.168.1.53:54189     192:8009               ESTABLISHED     32420
  TCP    192.168.1.53:56093     192:8009               ESTABLISHED     16008
  TCP    192.168.1.53:56094     192:8009               ESTABLISHED     16008
  TCP    192.168.1.53:56098     192:8009               ESTABLISHED     12244
  TCP    192.168.1.53:56522     192:8009               ESTABLISHED     32420
  TCP    192.168.1.53:56600     ec2-18-97-36-65:https  ESTABLISHED     32420
  TCP    192.168.1.53:57783     192:8008               ESTABLISHED     16008
  TCP    192.168.1.53:57784     192:8008               ESTABLISHED     16008
  TCP    192.168.1.53:57789     192:8008               ESTABLISHED     12244
  TCP    192.168.1.53:57790     192:8008               ESTABLISHED     12244
  TCP    192.168.1.53:60739     lb-140-82-113-26-iad:https  ESTABLISHED     32420
  TCP    192.168.1.53:60741     ec2-98-85-208-136:https  ESTABLISHED     30016
  TCP    192.168.1.53:60951     192:8009               ESTABLISHED     19924
  TCP    192.168.1.53:61178     ec2-18-97-36-44:https  ESTABLISHED     32420
```

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```
C:\Users\Utkar>netstat -r
===========================================================================
Interface List
  6...ac 19 8e e8 f5 f4 ......Microsoft Wi-Fi Direct Virtual Adapter
 10...ae 19 8e e8 f5 f3 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 19...b2 4a 34 58 0c c4 ......Intel(R) Wi-Fi 6E AX211 160MHz
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.1.1     192.168.1.53     35
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      192.168.1.0    255.255.255.0         On-link      192.168.1.53    291
     192.168.1.53  255.255.255.255         On-link      192.168.1.53    291
    192.168.1.255  255.255.255.255         On-link      192.168.1.53    291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link      192.168.1.53    291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link      192.168.1.53    291
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
 19    291 ::/0                     fe80::1
  1    331 ::1/128                  On-link
 19    291 2401:4900:1c6e:c2fe::/64 On-link
 19    291 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05/128
                                    On-link
 19    291 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4/128
                                    On-link
 19    291 fe80::/64                On-link
 19    291 fe80::948e:7197:14dc:d497/128
                                    On-link
  1    331 ff00::/8                 On-link
 19    291 ff00::/8                 On-link
===========================================================================
Persistent Routes:
```

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```
C:\Users\Utkar>netstat -e
Interface Statistics

                           Received            Sent

Bytes                      79068230       346668168
Unicast packets             2160390         1681470
Non-unicast packets           86334           10134
Discards                          0               0
Errors                            0               0
Unknown protocols                 0
```

Step 7 – ARP

```
C:\Users\Utkar>arp -a

Interface: 192.168.1.53 --- 0x13
  Internet Address      Physical Address      Type
  192.168.1.1           b0-8b-92-94-f9-f2     dynamic
  192.168.1.7           3c-c5-dd-6b-c6-37     dynamic
  192.168.1.11          78-b3-9f-e4-2e-95     dynamic
  192.168.1.33          30-68-93-28-04-18     dynamic
  192.168.1.49          50-bb-b5-f2-43-e0     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Step 8 – Hostname

```
C:\Users\Utkar>hostname
Utkarsh
```

Step 9– PathPing

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```
C:\Users\Utkar>pathping python-quiz.pages.dev

Tracing route to python-quiz.pages.dev [2606:4700:310c::ac42:2edd]
over a maximum of 30 hops:
  0  Utkarsh [2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05]
  1  2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2
  2     *        *        *
Computing statistics for 25 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           Utkarsh [2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05]
                              100/ 100 =100%   |
  1  ---     100/ 100 =100%     0/ 100 =  0%  2401:4900:1c6e:c2fe:b28b:92ff:fe94:f9f2

Trace complete.
```

Step 10 – Route Print

EXPERIMENT 2

```
C:\Users\Utkar>route print
===========================================================================
Interface List
  6...ac 19 8e e8 f5 f4 ......Microsoft Wi-Fi Direct Virtual Adapter
 10...ae 19 8e e8 f5 f3 ......Microsoft Wi-Fi Direct Virtual Adapter #2
 19...b2 4a 34 58 0c c4 ......Intel(R) Wi-Fi 6E AX211 160MHz
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.1.1    192.168.1.53     35
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      192.168.1.0    255.255.255.0         On-link      192.168.1.53    291
     192.168.1.53  255.255.255.255         On-link      192.168.1.53    291
    192.168.1.255  255.255.255.255         On-link      192.168.1.53    291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link      192.168.1.53    291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link      192.168.1.53    291
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
 19    291 ::/0                     fe80::1
  1    331 ::1/128                  On-link
 19    291 2401:4900:1c6e:c2fe::/64 On-link
 19    291 2401:4900:1c6e:c2fe:34f6:a7c0:a252:6a05/128
                                    On-link
 19    291 2401:4900:1c6e:c2fe:c0c5:a588:8649:88d4/128
                                    On-link
 19    291 fe80::/64                On-link
 19    291 fe80::948e:7197:14dc:d497/128
                                    On-link
  1    331 ff00::/8                 On-link
 19    291 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

Step 11 – Net Use

EXPERIMENT 2

```
C:\Users\Utkar>

C:\Users\Utkar>net use
New connections will be remembered.

There are no entries in the list.
```

Learning Outcome

After completing this experiment, I was able to:

- Understand the purpose and functionality of essential Windows networking commands.
- Test network connectivity and identify communication issues using Ping.
- Examine and manage TCP/IP configuration using IPConfig.
- Trace packet routes and identify intermediate network devices using Tracert.
- Resolve domain names and verify DNS services using NSLookup.
- Monitor network connections and protocol statistics using Netstat.
- Analyze ARP cache entries and understand IP-to-MAC address mapping.
- Identify the system hostname and network identity.
- Detect packet loss and network delays using PathPing.
- View routing information and shared network resources using Route and Net Use.
- Develop practical troubleshooting **skills for diagnosing common network problems.**